

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics COURSE CODE: CSA1504

COURSE OUTCOME COVERED

CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I **V.Raj kumar(192525208)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis

Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation
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Total Marks Awarded:

Assessment Tasks-1

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Screenshots:

Library Book Reservation - All Books

Book ID	Book Title	Author	Publisher	ISBN	Category	Copies Available	Status
B005	Echoes in Time	Sophie Turner	Chrono Publishing	978-4-56789-012-3	History	7	Unavailable
B004	Neon Cities	Carlos Mendez	Urban Edge	978-3-45678-901-2	Urban Fiction	2	Reserved
B003	Whispers of Wind	Lena Kim	Breeze Books	978-2-34567-890-1	Nature	8	Checked Out
B002	Quantum Dreams	Raj Patel	Nova Science	978-0-98765-432-1	Science	3	Unavailable
B001	The Silent Mountain	Elena Reed	Sunrise Press	978-1-23456-789-0	Fiction	5	Available
B006	White nights	Franz kafka	William publications	124824141	Fiction	3	Available

Library Book Reservation - Enter Reservation ID

Search

Total Fine Amounts
\$ 8.00

Total Reservations
5

Total Books
6

Total Students
5

Statuses

2 Returned

2 Confirmed

1 Overdue

0 Pending

All Reservations

Reservation ID	Student	Book	Reservation Date	Return Date	Fine Amount	Status
R005	S101 S102 S103	B001 B003 B004	05-Mar-2026	22-Mar-2026	\$ 0.50	Confirmed
R004	S101 S102	B002	05-Mar-2026	20-Mar-2026	\$ 3.25	Overdue

Library Book
Reservation

Dashboard

 Reservations

Books

+ Book

All Books

Statutes

Students

All Books

 ☐	Book ID ▾	Book Title ▾	Author ▾
	B005	Echoes in Time	Sophie Turner
	B004	Neon Cities	Carlos Mendez
... ☐	B003	Whispers of Wind	Lena Kim
	B002	Quantum Dreams	Raj Patel
	B001	The Silent Mountain	Elena Reed
	B006	White nights	Franz kafka

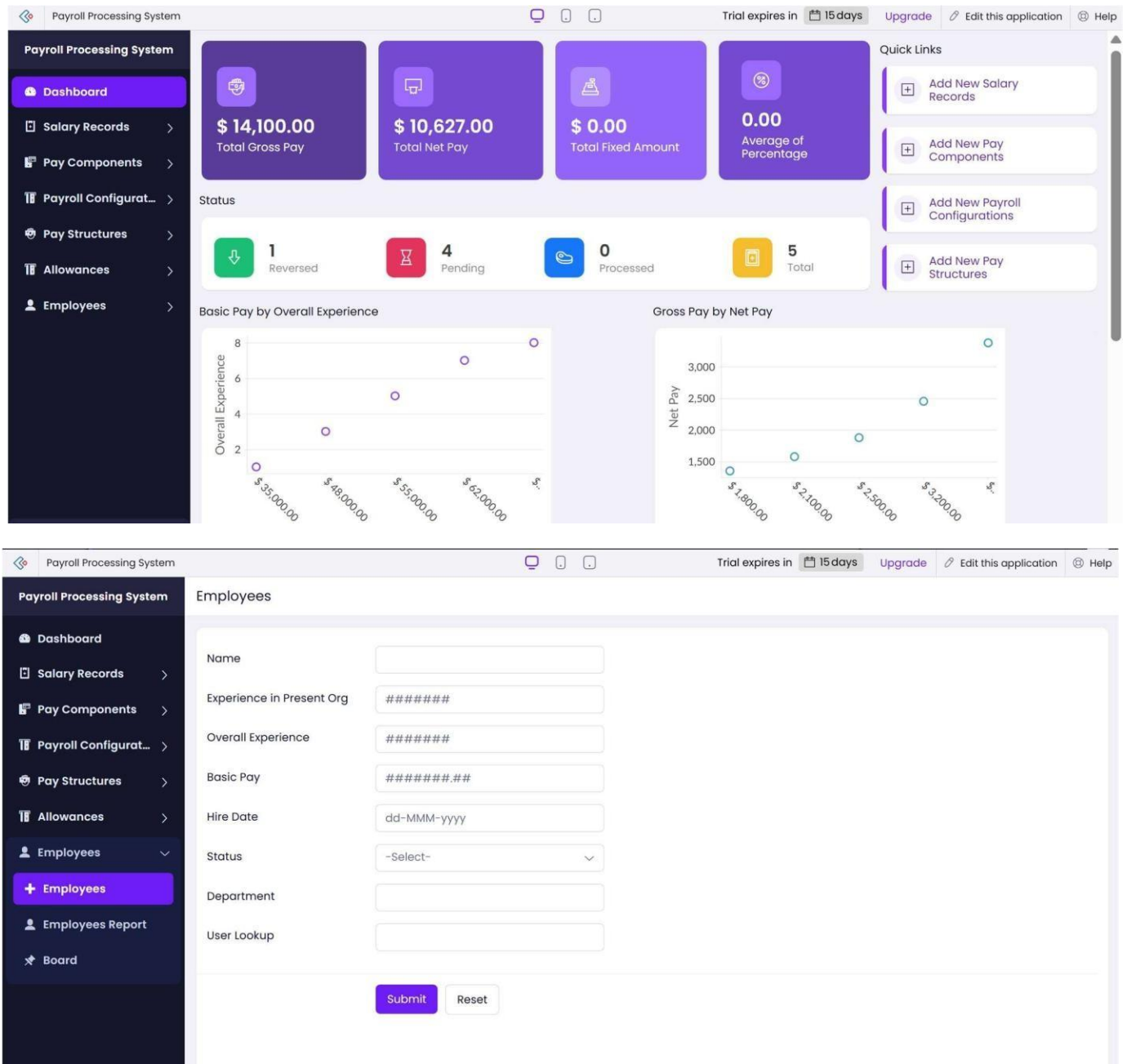
Overview

Book ID	B003
Book Title	Whispers of Wind
Author	Lena Kim
Publisher	Breeze Books
ISBN	978-2-34567-890-1
Category	Nature
Copies Available	8
Status	Checked Out



Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

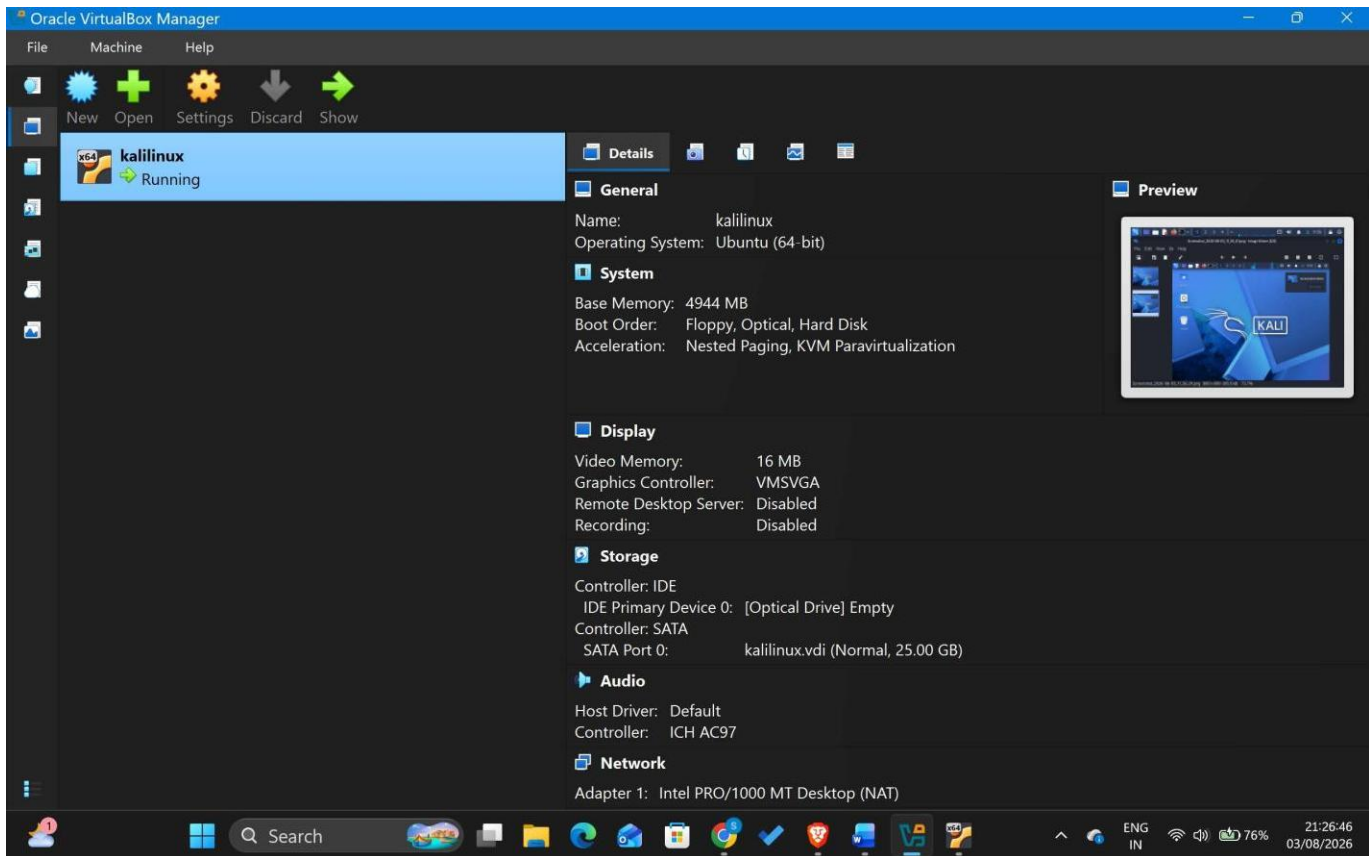
Screenshots:



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Screenshots:





Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Screenshots:

